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From Blockage to Breakthrough: Engineered Novel Blockage Blend for Controlled Fracture Growth

Marcus Chan Choon Hoong, Zaki Hussain A Alsandal, Azhar Alhasawi, and Ibrahim Albalawi, National Energy Services Reunited, Saudi Arabia; Anwar Husen Bin Akbar Ali, National Energy Services Reunited, Dubai

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Abstract

This study evaluates the performance of **Con-Fined Frac™ (CFF)** blend compared to a **Conventional Settle Frac (CSF)** proppant/sand system in hydraulic fracturing applications. The **CSF** blend consists of proppant and sand ranging from fine to coarse mesh sizes, designed to form a hexagonal spherical packing that creates artificial barriers to limit fracture growth, particularly in unwanted vertical directions. **The Con-Fined Frac™ (CFF)** blend drastically improves overall performance by incorporating degradable materials, including fibrous components and fine particulates; reducing the content of the conventional particulates blended in special fluid formulation. Fine particulates occupy voids between the **CSF** particles, enhancing sealing, while fibrous materials lock the blend together, reinforcing mechanical strength and integrity.

The **CFF™** fluid blend reduces the settling tendency and friction pressure of the total fluid system in the tubular and fracture face. Consequently conductive/effective fracture length is improved. The **CFF™** blend also results in an effective filter cake that reduces fluid loss and improves bridging across both small and large apertures, including natural fractures and high permeability stratigraphic layers away from the wellbore.

Laboratory evaluations, including bridging, leak-off, suspension, and degradation tests, demonstrate superior performance of **CFF™**. In bridging tests simulating a 2.5 mm far-field aperture, **CSF** exhibited immediate fluid bleed-off and poor particle distribution, whereas **CFF™** delayed leak-off by approximately 10 minutes, maintained particle suspension, and produced a spongy, well-bonded filter cake. Leak-off tests confirmed that **CFF™** limited fluid loss to about 35% versus 80% for **CSF**, maintaining stable pressure and effectively preventing fluid migration. Suspension tests showed that **CFF™** remained in stable suspension for over two hours, ensuring deeper and more uniform particle transport. Degradation tests using the high temperature blend indicated that **CFF™**'s degradable materials fully broke down within 43 hours at 200 °F, reducing near-wellbore and far-field proppant and conductivity damage. Using low temperature blend the degradation time can be reduced.

Overall, **CFF™** offers controlled fluid leak-off, effective bridging, sustained suspension stability, and complete degradation which provides a comprehensive solution for confining frac treatments in challenging formations. Its predictable fluid loss behavior is particularly valuable in low to high permeability reservoirs

and complex multi-phase flow scenarios. Optimal field performance requires proper laboratory evaluation, pumping design and fluid sequencing, emphasizing the integration of **CFF™** with engineering best practices. At the time of paper submission, one field frac treatment has been executed with great success in operation and well productivity response. A follow up paper would be submitted in future SPE technical conference.

Introduction

Over the past several decades, the oil and gas industry has experienced a marked shift in exploration and production focus. This change is primarily driven by the gradual depletion of easily accessible, high-productivity conventional reservoirs. Historically, these reservoirs—characterized by good porosity, high permeability, and favorable pressure conditions—allowed for economic hydrocarbon production with relatively straightforward completion and stimulation strategies. However, as these reserves have matured, operators are increasingly moving into more technically challenging and geologically complex environments to sustain production and meet global energy demand.

One significant challenge faced in these marginal reservoirs is the need to conduct hydraulic fracturing in formations located near water-bearing zones. Such zones may include underlying aquifers, overlying water-saturated layers, or laterally adjacent intervals with high water saturation. In these scenarios, uncontrolled fracture growth can result in direct communication with the water-bearing zone. This leads to water production, which reduces hydrocarbon output, increases water handling costs, and may cause long-term reservoir performance decline. In extreme cases, water breakthroughs can render a well uneconomical.

The complexity of the problem is heightened in heterogeneous formations where variations in lithology, mechanical properties, and in-situ stress can influence fracture propagation. In formations with low vertical stress contrast between the target pay and adjacent water-bearing layers, hydraulic fractures may grow uncontrollably into unwanted zones. Traditional fracturing methods, which focus on maximizing fracture conductivity and length, may not provide the level of fracture height control required in such sensitive environments. Therefore, specialized techniques are required to ensure stimulation efficiency while minimizing water production risks.

One of the proven solutions to address this challenge is the Settle Frac technique. This approach is specifically designed to limit fracture height growth by strategically utilizing proppant/sand settling behavior during the fracturing process. The principle is based on the fact that in a viscous fracturing fluid, proppant/sand particles gradually settle downward under gravity. By carefully designing pumping schedules, fluid rheology, and proppant concentrations, operators can encourage proppant to accumulate in the lower section of the fracture near the target pay zone.

In practice, the Settle Frac technique involves creating an initial fracture geometry, followed by the placement of proppant-laden slurry under conditions that promote rapid settling. As presented by [Barree et al. \(1995\)](#) and [Nitters et al. \(1996\)](#), as proppant/sand settles, it partially bridges or chokes the lower fracture area, effectively restricting further downward fracture growth into the water-bearing interval. Meanwhile, the upper sections of the fracture remain open, allowing continued upward and lateral fracture development within the hydrocarbon-bearing zone. The technique can be enhanced by varying the viscosity of the carrier fluid, adjusting pumping rates, and sequencing stages to optimize proppant placement.

[Valiulin et al. \(2019\)](#), [Hegazy et al. \(2021\)](#) and [Al-Yaqoubi \(2022\)](#) have presented successful cases of fracture height containment using Settle Frac technique in near proximity water bearing zone. [Garcia et al. \(2001\)](#) further provided supporting post-frac radioactive tracer survey data to validate this fracture height containment technique.

An effective Settle Frac technique is the use of a specialized proppant/sand blend (i.e. **Conventional Settle Frac (CSF)**) which normally consists of the following mesh sizes:

- 12/20 mesh
- 40/70 mesh
- 100 mesh

The **CSF** consists of larger and smaller size materials to create a hexagonal spherical packing, creating an artificial barrier and choking certain part of the fracture to limit fracture growth in unwanted directions (i.e. especially vertically downward).

The primary goal of this technological advancement is to create a diverter blend with improved ability to partially bridge or choke unwanted zones, thereby limiting fracture propagation into non-target intervals. The newly developed blend is designed to deliver superior control of fluid leak-off, enhanced bridging efficiency, and excellent suspension stability, offering clear advantages over the **CSF** commonly applied in Settle Frac treatments. The next section details the novel diverter blend product.

Innovative Improved Settle Frac (CFF™)

Historically, the Settle Frac technique has shown mixed results due to its strong dependence on factors such as stress contrast and permeability differences between the pay zone and the adjacent water-bearing zone. To enhance its effectiveness, a novel approach—termed the **Con-Fined Frac™ (CFF)** was developed. This method incorporates the proppant/sand blend together with degradable materials (i.e. fibrous materials and fine particulates). Table 1 summarizes the **CFF™** features.

Table 1—Summary of CFF™ features

	Description
Application	• Primarily used as the specialized blend in Settle Frac to create artificial barrier and choking certain part of fracture to limit fracture growth
CFF™ Blend	• CFF™ Blend consists of the following ○ CSF (consisting of larger and smaller proppant/sand at lower quantity) ○ Absorbent particulates ○ Fibrous material
Working Principle	• CSF blend serves as the main structure to create artificial barrier and to choke of certain part of the fracture • Fine-size particulates will occupy the void spaces between the CSF blend improve the sealing mechanism • Fibrous materials lock the CSF blend and the fine-size particulates reinforcing mechanical strength of the blend integrity • The blend creates effective filter cake, reducing fluid loss through porous media and bridging capability in small and large aperture
Field Mixing & Treatment Execution	• The diverter blend can be mixed on the fly using frac blender or batch mixed. • The CFF™ blend is pumped at low concentration with the volume 5 to 20% of the treatment proppant volume

Experimental Study

An experiment study was performed to evaluate the effectiveness of **CFF** by comparing it with the benchmark **CSF** based on the following tests:

- Bridging test
- Leak-off test
- Suspension test
- Degradation test

Bridging Test

Bridging test was done using a specialized apparatus as shown in Fig. 1. The bridging test assesses a diverter material's capability to form an effective seal across fracture widths under simulated downhole conditions. This key laboratory evaluation determines the diverter's ability to:

- Rapidly create stable bridges at target fracture widths
- Withstand differential pressures experienced during fracturing
- Maintain structural integrity until intentional degradation or removal

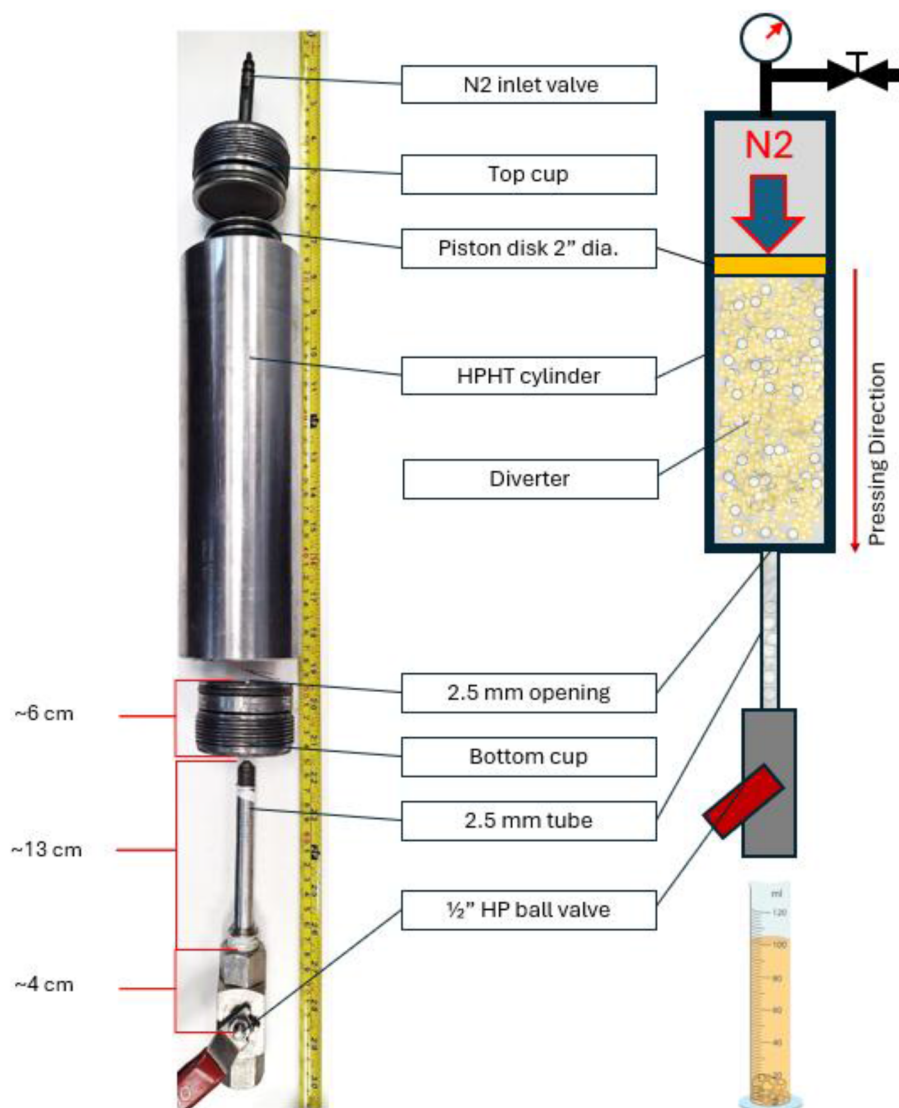


Figure 1—Bridging test apparatus

Bridging test was done using a specialized apparatus as shown in Fig. 1. The apparatus features a pressurized chamber equipped with a top-mounted N₂ inlet valve, which drives a disk piston downward to apply controlled force on the diverter slurry. At the base, a precisely machined tunnel measuring 2.5 mm (~0.1 in) diameter and 230 mm in length simulates challenging far-field fracture dimensions. Gomaa et al. (2016) recommended the slot width near to 0.1 in to properly assess diversion in far field.

Pressure is monitored using a digital gauge, while a digital controller preheats the chamber to 190 °F. Testing begins by preheating the closed cell to approximately 70% of the target temperature for about 10 minutes before introducing the diverter. With the bottom valve closed, the system is stabilized at test temperature, after which the 2.5 mm aperture is opened to start the experiment. Pressure is increased in increments of 100 PSI every 2.5 minutes, and expelled fluid is collected in a graduated cylinder to determine the leak-off rate (LOR).

The primary performance criterion is maintaining 500 PSI for one hour without complete leak-off (>80% volume loss) or sudden blowout. After testing, the filter cake is inspected for height, texture, wetness, and particle bonding, while any migrated solids in the tunnel are analyzed. LOR is calculated as $(\text{accumulated volume} \div \text{total diverter volume}) \times 100\%$, with values over 80% indicating failure.

Secondary analysis involves monitoring pressure decay after closing the inlet valve—rapid decline suggests high porosity, whereas stable pressure reflects effective bridging. This evaluation not only determines pass/fail status but also provides diagnostic insights to refine diverter formulations and forecast field performance in far-field fracture scenarios.

Fig. 2 chart above compares bridging test results for the CSF and CFF™. In the CSF test, fluid bleed-off began almost immediately, rapidly reaching maximum leak-off. The system failed to build pressure beyond the initial 100 PSI, indicating that liquid and some solids flowed freely through the 2.5 mm tunnel. This behavior points to rapid particle settling, which can cause inefficient pumping and poor particle placement within the fracture network.

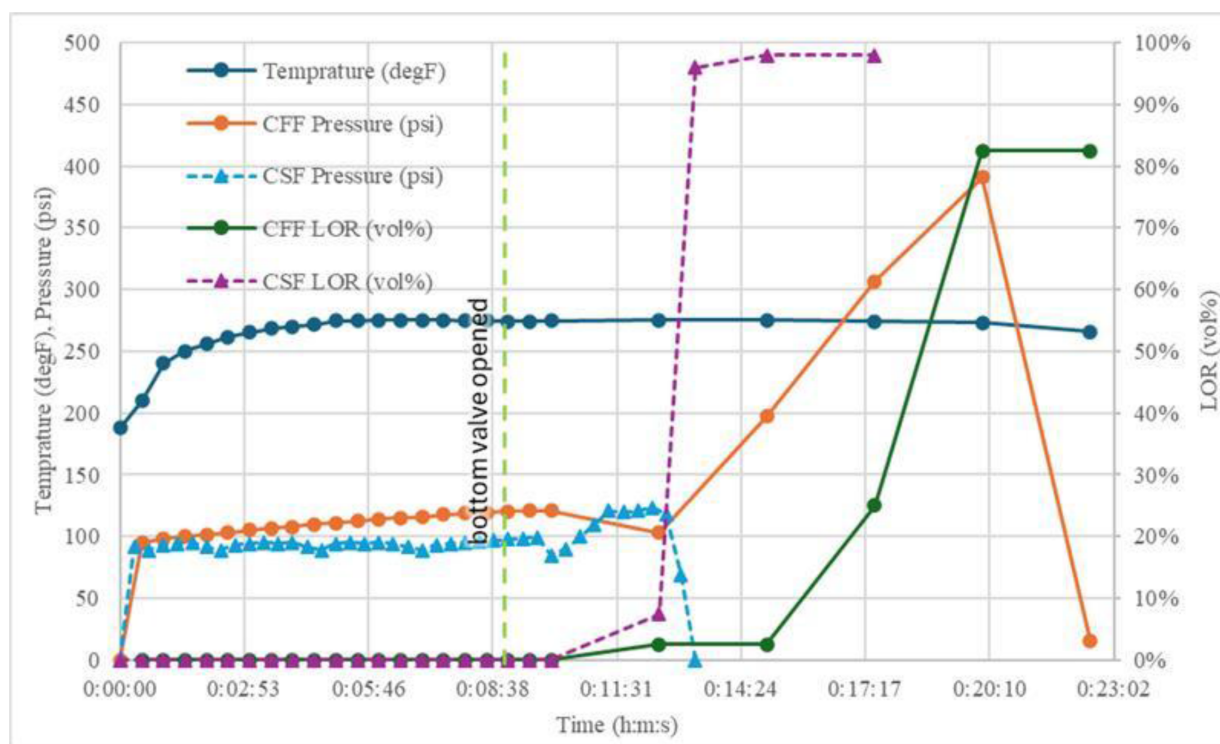


Figure 2—Bridging test comparison between CFF™ vs CSF

In contrast, under identical conditions, the CFF™ exhibited a significantly longer plugging time, with about additional 10-minute before substantial leak-off occurred. This delay indicates that the CFF™ maintained suspension and transport of particles more effectively than CSF. Additionally, the CFF™'s leak-off rate showed a more gradual and controlled decline during the injection period, suggesting enhanced flow control and deeper diverter penetration into fracture zones—areas where CSF would be less effective in delivering material.

Upon opening the HPHT apparatus, further evaluation was conducted by inspecting the 2.5 mm hole, the corresponding tunnel, and the bottom valve. In the **CSF** test, the filter cake was found to be brittle, poorly bonded, and dry (i.e. shown in Fig. 3)). Particle distribution was uneven, with finer particles settled at the top and coarser particles concentrated at the bottom.

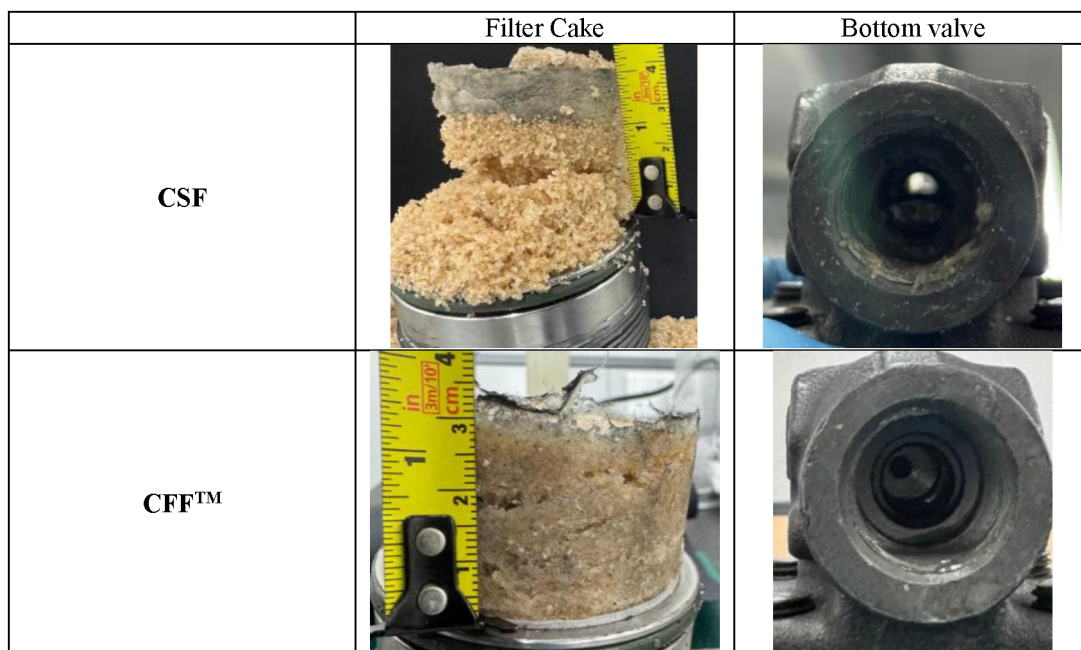


Figure 3—After-test evaluation photos for bridging test using 2.5 mm opening for CSF & CFF

In contrast, the **CFF™** test produced a spongy, soft, and well-bonded filter cake with uniform particle distribution throughout (i.e. shown in Fig. 3). This even dispersion, combined with strong particle bonding, indicates that a cohesive filter cake with well-distributed particles can deliver superior bridging capability and more controlled leak-off rate (LOR) performance.

Leak-Off Test

The leak-off test is a critical evaluation to assess the diverter's ability to control fluid loss into far-field fractures. This test measures the diverter's sealing capacity by analyzing fluid filtration rates under simulated downhole conditions (pressure, temperature, and fracture geometry). Effective leak-off control ensures that the diverter temporarily blocks fractures taking the fluid.

The leak-off test evaluates a diverter's ability to form an effective seal inside fractures under simulated downhole conditions. This test measures the rate of fluid filtration through a diverter-packed fracture, providing key insights into:

The leak-off test provides critical insights into filter cake formation dynamics, including:

- Particle distribution - Uniformity of deposited solids across the fracture face
- Filter cake height - Measured thickness of the consolidated particle layer
- Structural integrity - Particle-to-particle bonding and pore structure stability

Leak-Off Rate (LOR) interpretations are as follows:

1. Excessive Leak-Off (Fast Rate):
 - Indicates poor particle suspension and bridging capability

- Leads to inadequate far-field diversion due to:
 - Premature particle settling
 - Insufficient fracture face coverage
 - Increases screen-out risk from uncontrolled fluid loss
2. Insufficient Leak-Off (Slow Rate):
- Suggests over-compacted filter cake formation
 - Creates potential conductivity impairment through:
 - Excessive cake thickness
 - Reduced permeability in near-fracture zone
 - May limit subsequent fluid/proppant placement
3. Optimal Leak-Off Range:
- Balanced rate that achieves:
 - Effective fracture face sealing
 - Maintained near-fracture conductivity
 - Progressive (not instantaneous) cake buildup

A controlled LOR can be the key to empowering far-field diversions because it regulates fluid loss into the formation, ensuring that the carrying fluid retains enough pressure and velocity to transport diverter material deep into the desired zones rather than prematurely depositing it near the wellbore.

This testing plan was designed using Fig. 4 set-up, to evaluate the impact of Optimal LOR on both CSF and CFF™ at different formation leak-off conditions. Two distinct scenarios were selected, differentiated only by the filtering media, to simulate representative formation types. The first scenario utilized a defined permeability synthetic ceramic (775 mD) to represent a tight, low-permeability formation with restricted fluid loss. The second scenario employed filter paper (11 µm) supported by wire mesh #325 (45 µm) to represent a high-permeability formation with larger pore throats and higher fluid loss potential. This dual-media setup allowed for a controlled investigation into how LOR influences diverter transport and placement in contrasting leak-off environments. 775 mD was the lowest permeability available and it can be tested on lower permeability range in next development. By testing the new product under both conditions and comparing its performance to established market products, the study aims to assess its ability to maintain an optimal LOR, prevent premature near-wellbore bridging, and promote effective far-field fracture coverage. The approach provides a robust and technically relevant framework for validating product performance across a realistic range of reservoir characteristics.

The LOR test is conducted using a specially modified HPHT cell designed to evaluate far-field diversion performance under simulated downhole conditions. The apparatus consists of a pressurized chamber with a top-mounted N₂ inlet valve that drives a disk piston downward, applying controlled force to the diverter slurry. The filtering media is positioned at the bottom which can be ceramic disk or filter paper supported by wire mesh. This represents challenging far-field fracture conditions. The bottom cup with open hole 2.5 mm (~0.1 inch). During testing, pressure is monitored via digital gauge while temperature is preheated at 190°F using a digital controller.

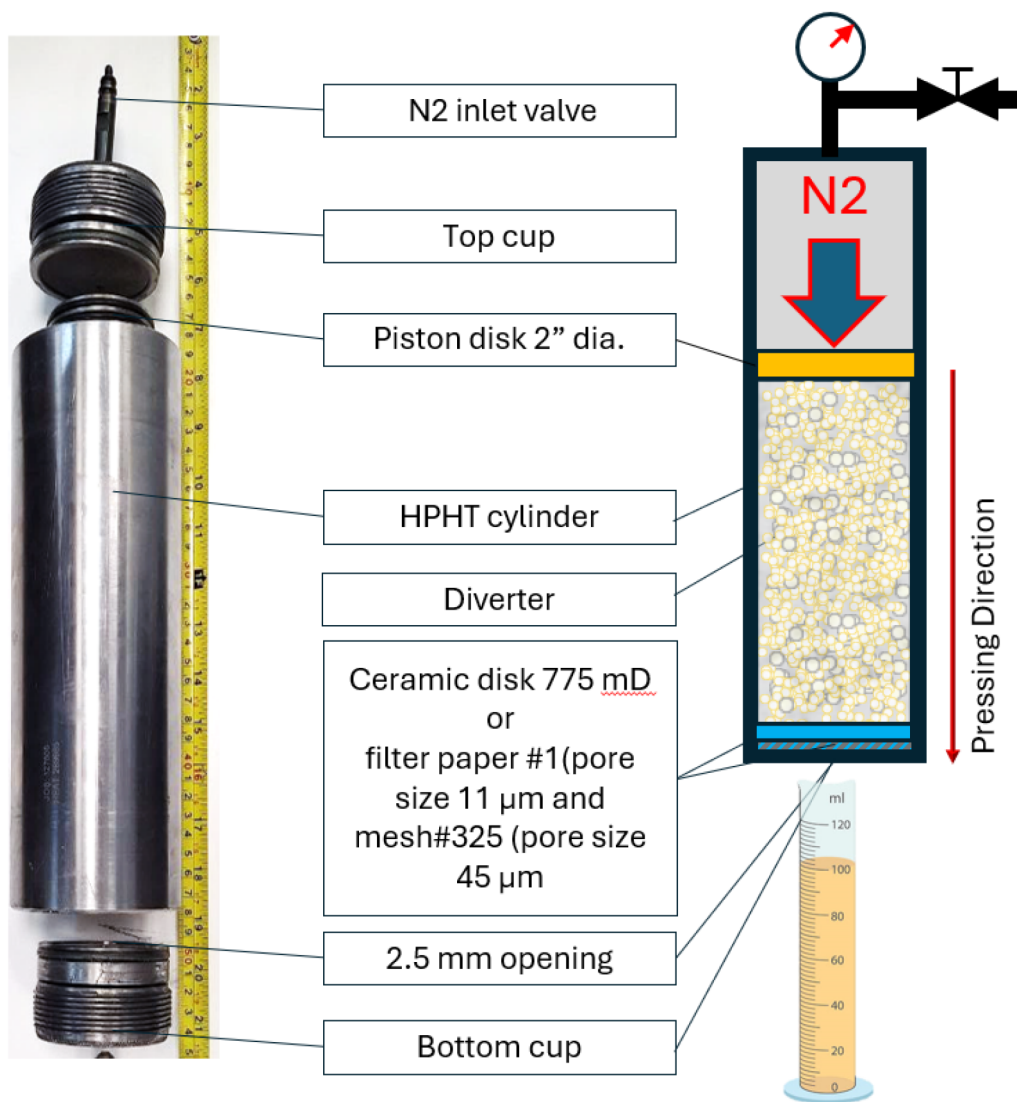


Figure 4—Leak-off test apparatus set-up

The experimental procedure starts by preheating the closed cell to 70% of the target temperature—according to the setpoint—for approximately 10 minutes before introducing the blend. With the bottom stem valve initially closed, the system is stabilized at the test temperature, after which the valve is opened to begin the experiment. Pressure is then increased in 100 PSI increments every 2.5 minutes, while expelled fluid is collected in a graduated cylinder to measure the leak-off rate (LOR).

The primary performance criterion is the ability to maintain 500 PSI for roughly one hour without experiencing complete leak-off (>80% volume loss) or sudden blowout. After the test, the filter cake is examined for key physical characteristics, including height, sponginess, moisture content, and particle bonding. The LOR is calculated as $(\text{accumulated volume} \div \text{total diverter volume}) \times 100\%$, with values above 80% considered a failure. Unlike the bridging test, only liquid phase is expected to be collected during the filtration experiment.

A secondary analysis of pressure decay after closing the inlet valve provides additional insight into filter cake quality—rapid pressure loss indicates high porosity, while stable pressure suggests effective bridging. This thorough evaluation not only confirms pass or fail status but also offers valuable diagnostic data for refining diverter formulations and predicting their performance in far-field fracture applications.

Chart from Fig. 5 indicated that the CSF demonstrated a clear correlation between leak-off rate (LOR) and rising pressure, with LOR peaking at approximately 80% as pressure neared 500 PSI. Once the inlet valve

was closed, pressure declined steadily. In contrast, the **CFF™** exhibited a much slower leak-off response, maintaining better control with LOR limited to about 35%. After the inlet valve was closed, pressure in the CFF test remained stable, indicating that the filter cake effectively sealed the trapped liquid phase while also preventing gas migration through it.

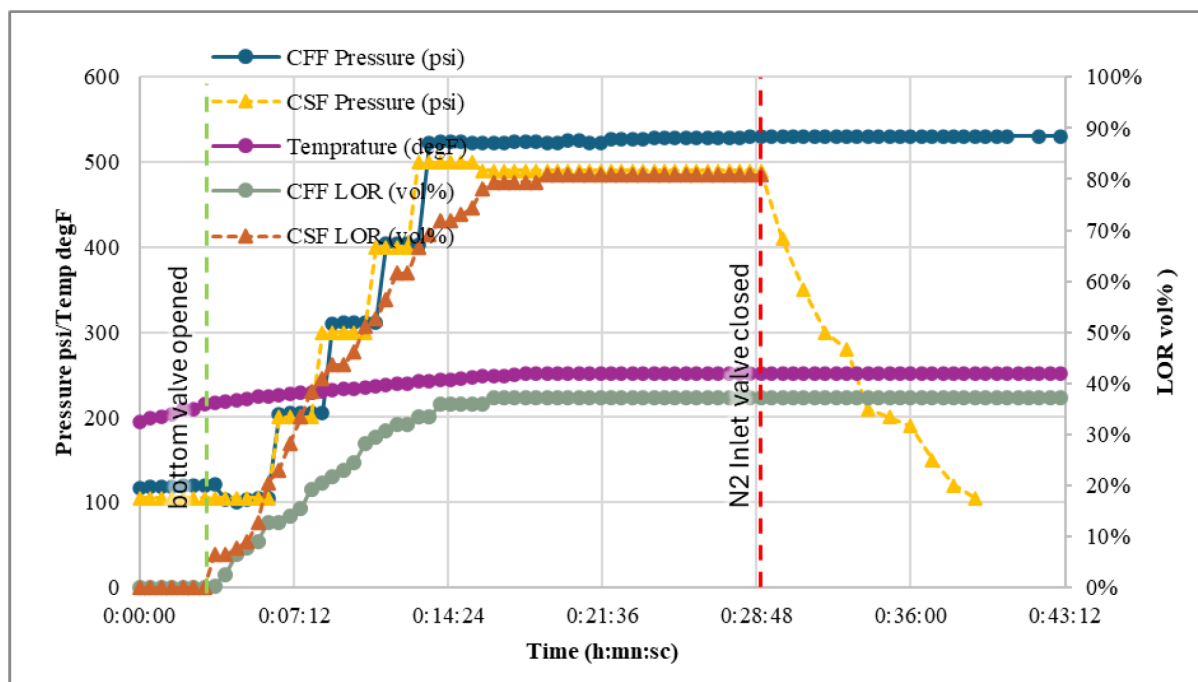


Figure 5—Leak-off test using 11 μ m filtering media for CSF and CFF™ at 250°F and 500 PSI

Fig. 6 illustrates the filter cakes obtained from leak-off tests conducted using 11 μ m filtering media at 250°F and 500 psi. In the **CSF** test, the resulting filter cake showed clear evidence of poor particle bonding. The side-view image highlighted non-uniform particle distribution, with fine particles migrating and concentrating at the upper layer while coarse particles settled predominantly at the bottom. This segregation reflected weak structural integrity of the filter cake. The LOR for **CSF** was approximately 80%, suggesting that only about 20% of the liquid remained trapped. However, this retention was attributed primarily to a viscous film created by the gelling agent rather than effective solids bridging, thus indicating minimal contribution from the solid additives in controlling fluid loss.

By contrast, the **CFF™** formulation produced a markedly different outcome. The filter cake was soft, spongy, and well-bonded, with evenly distributed particles throughout the structure. This more cohesive and homogeneous arrangement provided significantly improved leak-off resistance. Consequently, the LOR decreased from around 80% to approximately 35%, clearly demonstrating the superior performance of the enhanced **CFF™** formulation.

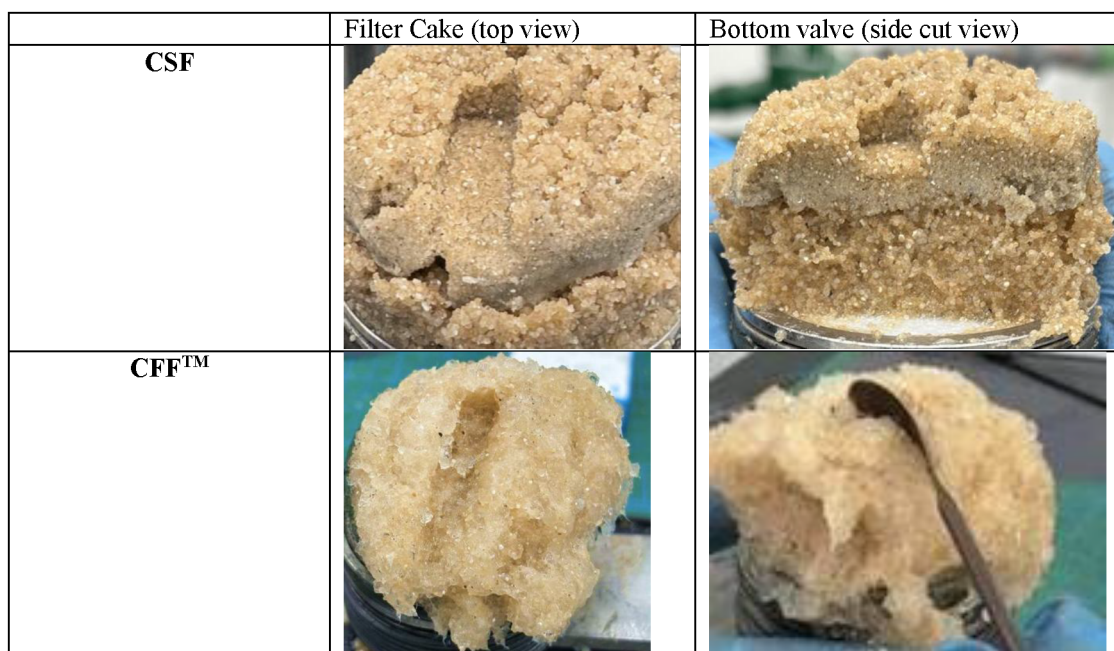


Figure 6—After-test evaluation photos for leak-off test using 11 μ m filtering media for CSF and CFF™ at 250 degF and 500 PSI

Suspension test

The suspension test as conducted in Fig. 7, evaluates the ability of the blend on particle settling, which is critical for maintaining uniform solid distribution during treatment. Given the high solid content in carrier fluids, effective suspension is essential to prevent premature settling, which can lead to operational challenges. Poor suspension may result in uneven diverter deployment, reducing its effectiveness in bridging and sealing far-field fractures. Key challenges associated with solid settling include:

- Non-uniform particle distribution – Leading to inconsistent fracture coverage and reduced diversion efficiency.
- Deployment difficulties – Hindering the delivery of a homogeneous diverter slurry into the formation.
- Reduced fracture conductivity – Settled solids may form localized plugs rather than a dispersed barrier, impairing well productivity.
- Optimizing suspension properties ensures better diverter performance by enhancing particle transport and placement in the fracture network.

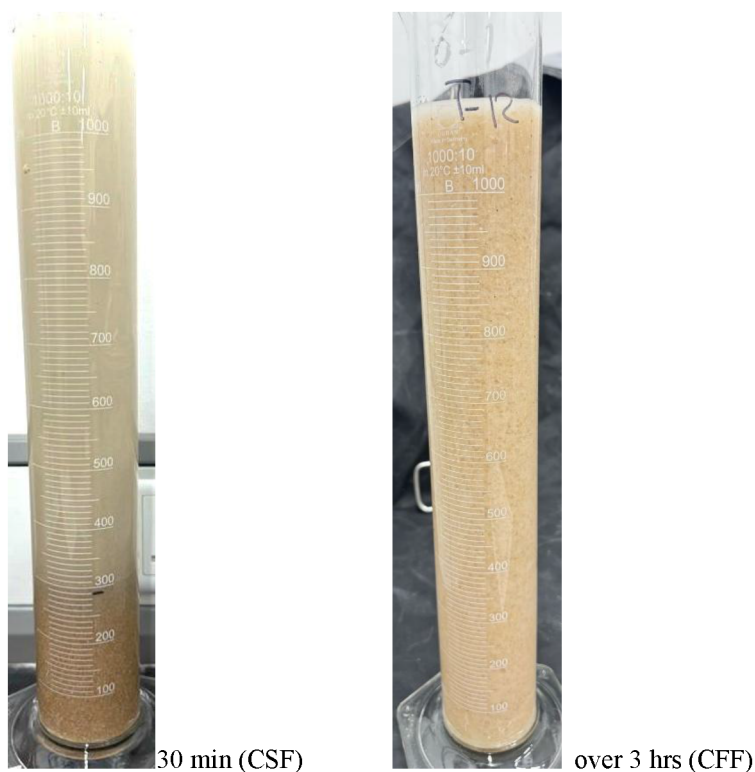


Figure 7—Suspension test at room temperature

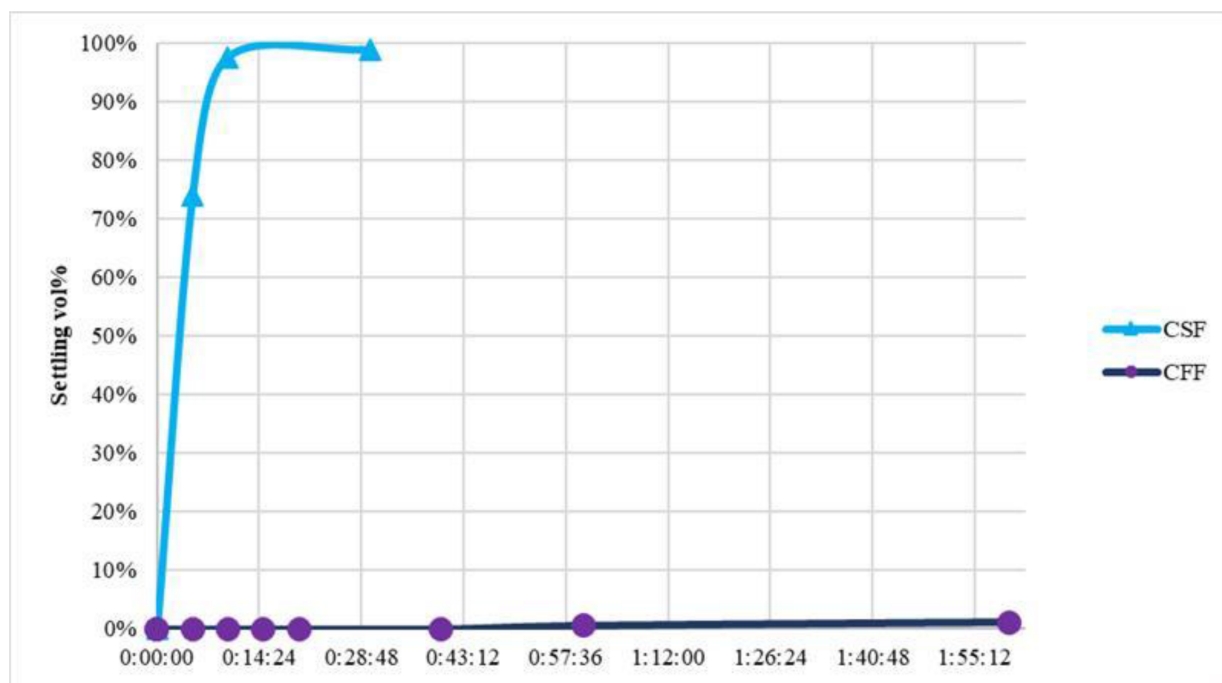


Figure 8—Suspension test chart

The settling tests revealed a clear difference in suspension behavior between the **CSF** and the **CFFTM** formulation. The **CSF** settled completely within just 10 minutes coarse particles rapidly dropping to the bottom and fine particles dispersing unevenly in the fluid. Such rapid settling highlights the limited stability of **CSF**, which significantly reduces its ability to maintain a uniform particle distribution during pumping

operations. As a result, its effectiveness in achieving consistent solids transport and placement deep into the fracture system is greatly compromised.

In contrast, the **CFF™** formulation displayed remarkable suspension stability. Even after more than two hours under static conditions, no visible settling was observed, indicating that the particles remained evenly distributed throughout the carrier fluid. This prolonged stability ensures that diverter particles are carried consistently with the fluid flow, minimizing premature settling near the wellbore and maximizing the potential for deeper transport.

Maintaining a homogeneous suspension is especially valuable in high-leak-off formations or complex fracture geometries, where effective far-field diversion is critical. The stable suspension of **CFF™** promotes more uniform placement, enabling efficient sealing of targeted fracture zones and improving overall fracture coverage, which ultimately contributes to better stimulation effectiveness and production outcomes.

Degradation Test

The key advantage of using the **CFF™** formulation over the **CSF** lies in its optimized design, which relies on a reduced quantity of proppant or sand in combination with degradable components such as fibrous materials and fine particulates. This tailored blend not only lowers the overall proppant requirement but also enhances placement efficiency, particularly in high-permeability formations. By partially substituting sand with degradable materials, **CFF™** minimizes near-wellbore conductivity, which is beneficial for promoting controlled fluid diversion into deeper sections of the reservoir.

To assess the degradability of these materials, a dedicated test was conducted using broken gel as the fluid medium as shown in Table 2. The results (i.e. Fig. 9) confirmed that the degradable additives in the **CFF™** formulation underwent complete degradation within 43 hours at 200°F. This ensures that no long-term residue or blockage remains in the formation, thereby mitigating potential formation damage and validating the clean, temporary nature of the diverter system.

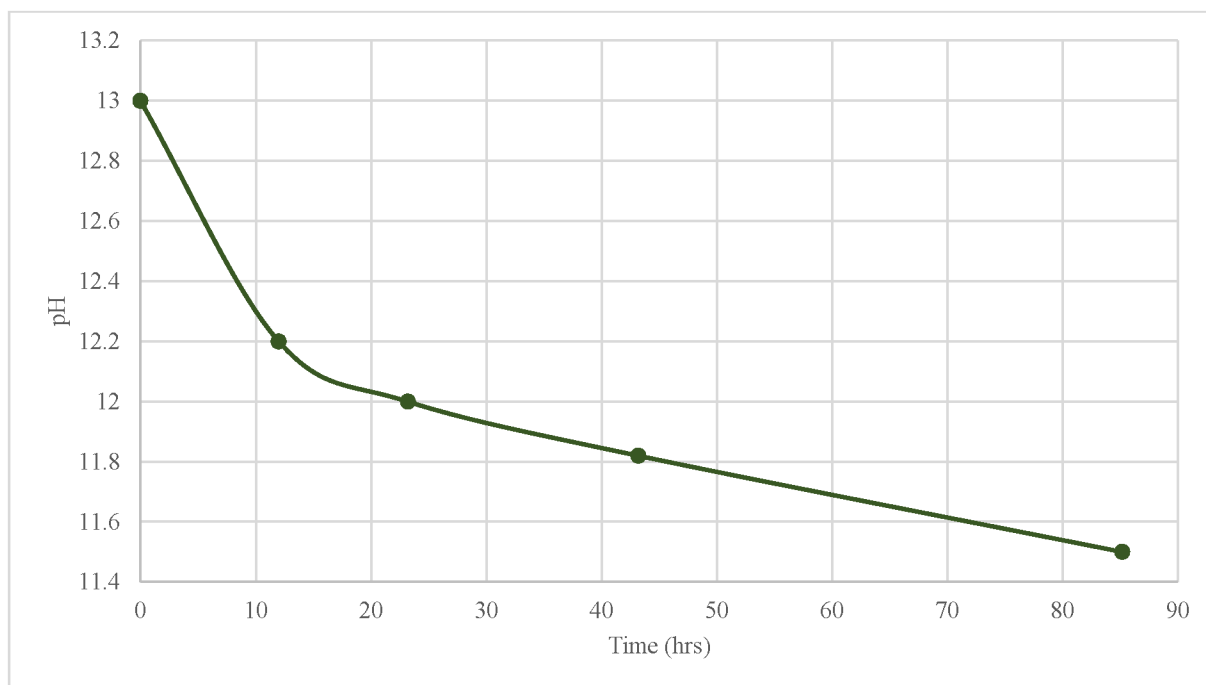
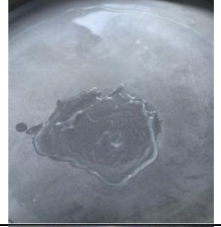


Figure 9—pH vs time chart for degradation progress at 200°F

Table 2—Degradation at 200°F

#	Time (hrs)	pH	Aged sample photo	Filter sample using mesh# 200
1	Start	13		
2	12	12.2		
3	23	12.0		
4	43	11.8		

The results showed that full degradation took place in less than 43 hours at test temperature of 200°F.

Conclusion

The following concludes the key points discussed in this publication:

- A specialized proppant/sand blend (i.e. **CSF**) consists smaller to bigger mesh sizes of proppant/sand are being used to create a hexagonal spherical packing, creating an artificial barrier and choking certain part of the fracture to limit fracture growth in unwanted directions (i.e. especially vertically downward).
- An improved blend (i.e. **CFF™**) consists of **CSF** and degradable materials (i.e. fibrous materials and fine particulates) are developed to enhance the blend performance in Settle Frac treatment.
 - Fine-size particulates will occupy the void spaces between the **CSF** blend improve the sealing mechanism
 - Fibrous materials lock the **CSF** blend and the fine-size particulates reinforcing mechanical strength of the blend integrity
 - The blend creates effective filter cake, reducing fluid loss through porous media and bridging capability in small and large aperture

- Laboratory results and evaluations indicate that the **CFF™** offers substantial advantages over the **CSF** in tackling critical challenges encountered during Settle Frac treatment
- **Bridging Test:**
 - The bridging test, conducted to simulate far-field conditions with a 2.5 mm opening, demonstrated superior performance for **CFF™**.
 - **CSF** exhibited immediate fluid bleed-off and could not build pressure above 100 PSI, indicating free flow of liquid and some solids. This reflects rapid particle settling, resulting in inefficient pumping and uneven particle distribution.
 - In contrast, **CFF™** showed a much longer plugging time, with approximately a 10-minute delay before significant leak-off, indicating effective suspension and transport of particles. Its leak-off rate also declined more gradually, suggesting improved control and deeper fracture penetration.
 - Post-test inspection revealed that **CSF** formed a brittle, unbonded, and dry filter cake with poor particle distribution, where the finest particles settled at the top and coarsest at the bottom.
 - **CFF**, on the other hand, produced a spongy, soft, well-bonded filter cake with uniform particle distribution, directly contributing to its superior bridging performance and controlled leak-off.
- **Leak-Off Test:**
 - The leak-off test assessed the ability to control fluid loss through the filtering medium simulating formation permeability.
 - **CSF** exhibited a direct correlation between LOR and increasing pressure, reaching about 80% at 500 PSI. After closing the inlet valve, pressure steadily declined. The remaining 20% of liquid was likely retained by the viscous gelling agent rather than effective solids bridging, indicating minimal resistance from the solid additives.
 - **CFF™** displayed a much slower leak-off response, limiting LOR to approximately 35%, and maintained stable pressure after the inlet valve was closed. This confirms that the **CFF** filter cake effectively sealed the trapped liquid and prevented gas migration.
 - The filter cake formed by **CSF** was unbonded with poor particle distribution, consistent with observations from the bridging test.
 - In contrast, **CFF™** produced a soft, spongy, well-bonded filter cake with uniform particle distribution, reducing LOR from 80% (**CSF**) to roughly 35%. This demonstrates **CFF™**'s superior performance in controlling fluid loss.
- **Suspension Efficiency:**
 - The suspension test evaluated the settling behavior of **CSF** and **CFF™** at room temperature.
 - **CSF** fully settled within 10 minutes.
 - **CFF™** remained in a highly stable suspension for over 2 hours with no visible settling, indicating it maintains a uniform particle distribution for a much longer period, ensuring more consistent transport of solids deep into the fracture network.
 - This prolonged suspension stability, improving far-field diversion
- **Degradation Test:**
 - The **CFF™** offers an advantage over **CSF** by using less proppant or sand combined with degradable materials such as fibrous components and fine particulates. In high-permeability formations, this reduces near-wellbore proppant volume and conductivity. Degradation tests showed these materials fully degrade within 43 hours at 200 °F.

In conclusion, the laboratory evaluations consistently demonstrate that the **CFF™** formulation delivers superior performance compared to the **CSF**. **CFF™** offers controlled fluid leak-off, strong bridging capability, excellent suspension stability, and complete degradability, making it a comprehensive and reliable solution for Settle Frac treatments, particularly in challenging formations. Its ability to provide a stable and predictable fluid loss coefficient is highly advantageous in medium- to high-permeability reservoirs as well as under multi-phase flow conditions.

The effectiveness of Settle Frac treatment is largely attributed to the diverter blend's capacity to partially bridge or choke undesired zones, thereby restricting fracture propagation in non-target directions. This mechanism helps control fracture growth within the desired pay zone while minimizing the risk of extending fractures into unwanted intervals, such as water-bearing layers. By ensuring more controlled fracture placement, **CFF™** enhances the efficiency of stimulation and improves overall production outcomes. The experimental findings further validate that **CFF™** is a more advanced and dependable diverter blend for Settle Frac applications compared to the conventional formulations commonly used in fracturing operations. Its balanced combination of performance and degradability positions it as a superior choice for optimizing fracture treatments in complex reservoir environments.

Abbreviations or Units

%	– Percentage
μm	– Micrometer (unit of length)
°F	– Degrees Fahrenheit (unit of temperature)
CFF	– Con-Fined Frac™
CSF	– Conventional Settle Frac
HPHT	– High-Pressure & High-Temperature
LOR	– Leak-Off Rate
mD	– Milli-Darcy
mm	– Millimeter (unit of length)
N ₂	– Nitrogen
PSI	– Pounds Per Square Inch (unit of pressure)

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